

Tuning in to black holes: is it a bird, is it a plane, or is it just a calibration error?

Impact of detector calibration accuracy on black hole spectroscopy [submitted]

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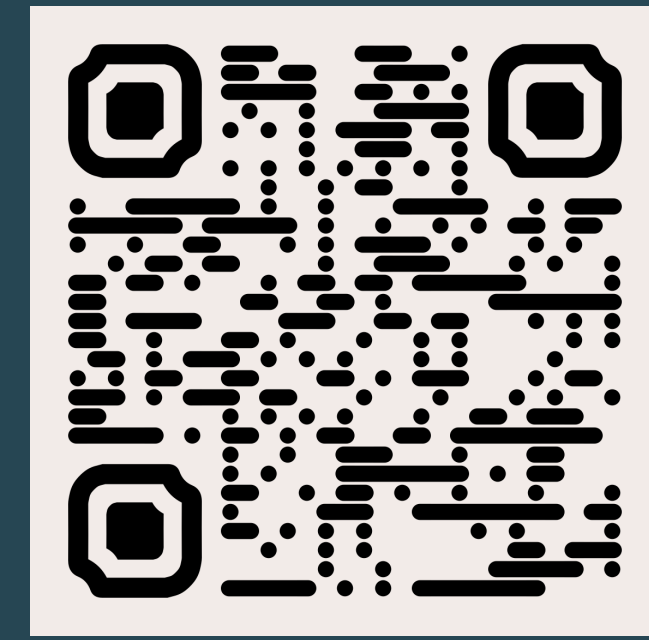
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Is that a non-GR effect or is that just a calibration error?

- GW detectors output dimensionless strain data, which is used for astrophysical studies
- Strain data is reconstructed by modelling the detector's response in a calibration procedure
- Imperfect modelling may result in miscalibrated strain data

'Inject' signal into a detector frame

ACCURATELY
CALIBRATED

MISCALIBRATED

Estimate BH mass and spin (M and χ)
using BH spectroscopy

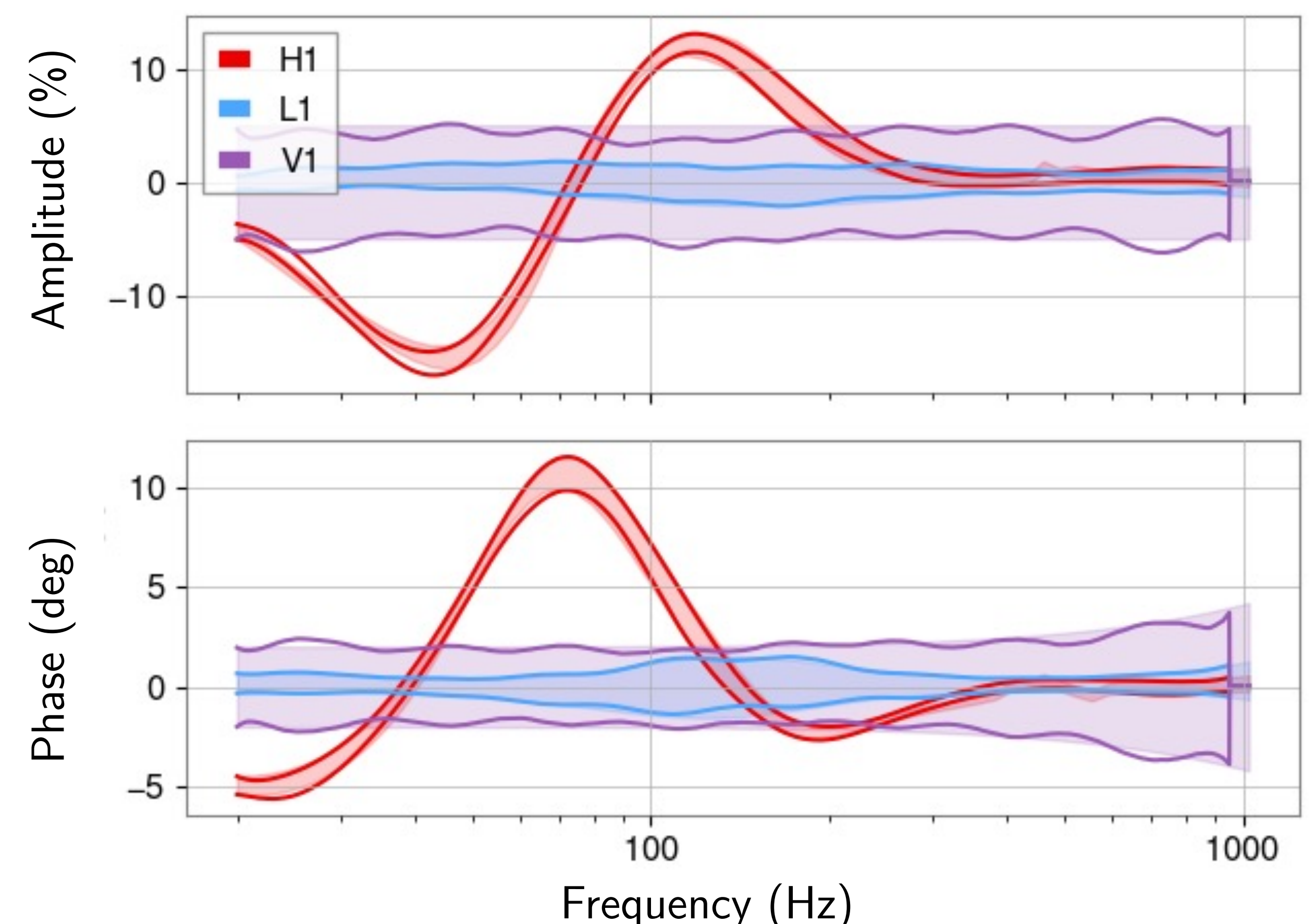
ACCURATELY
CALIBRATED

MISCALIBRATED

Compare to true BH properties

Will our
astrophysical
analyses be
biased? Let's
look at the
parameter
estimation!

Calibration error envelope – O4b aLIGO S240925n (it's really happening)

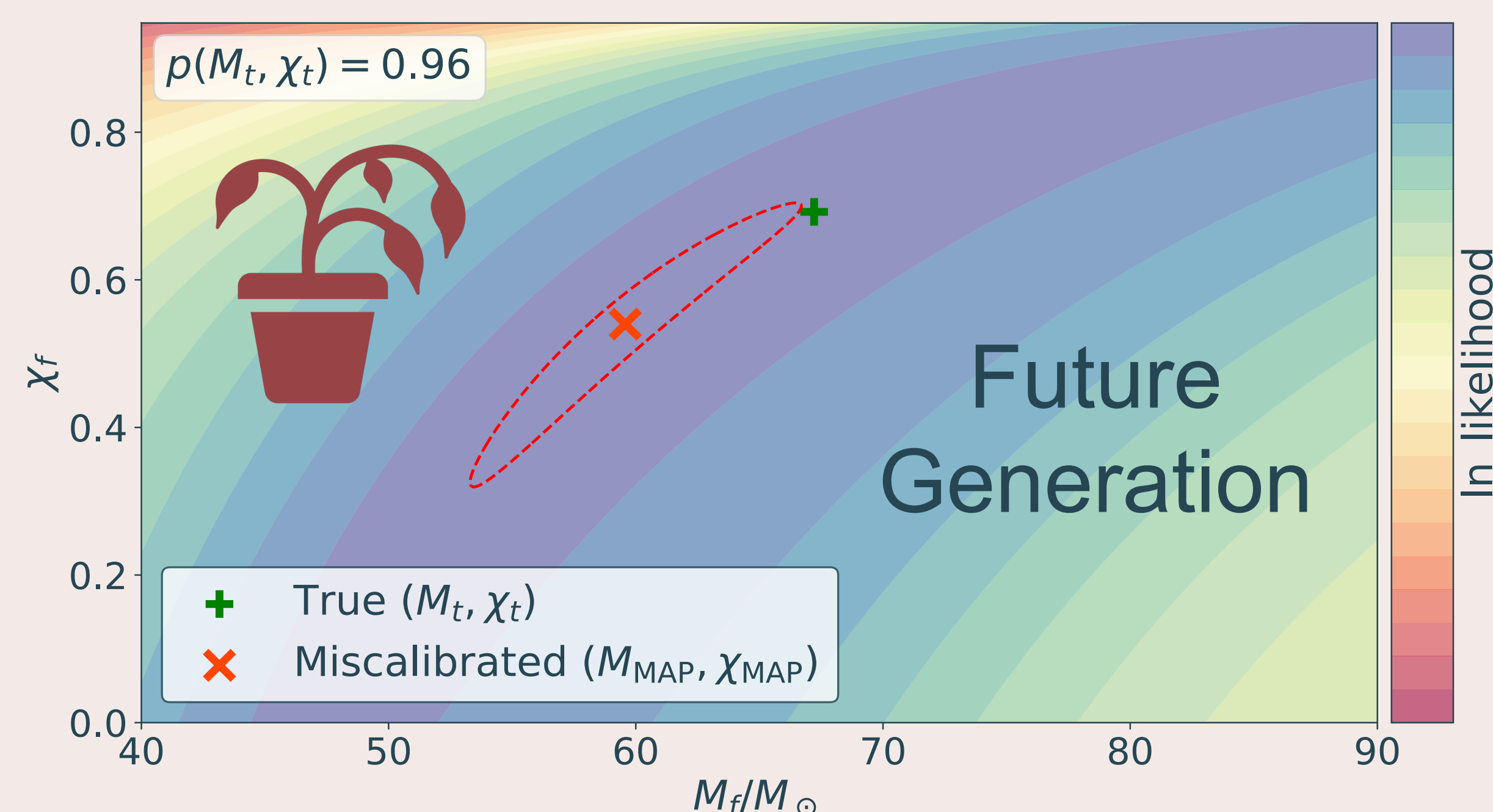
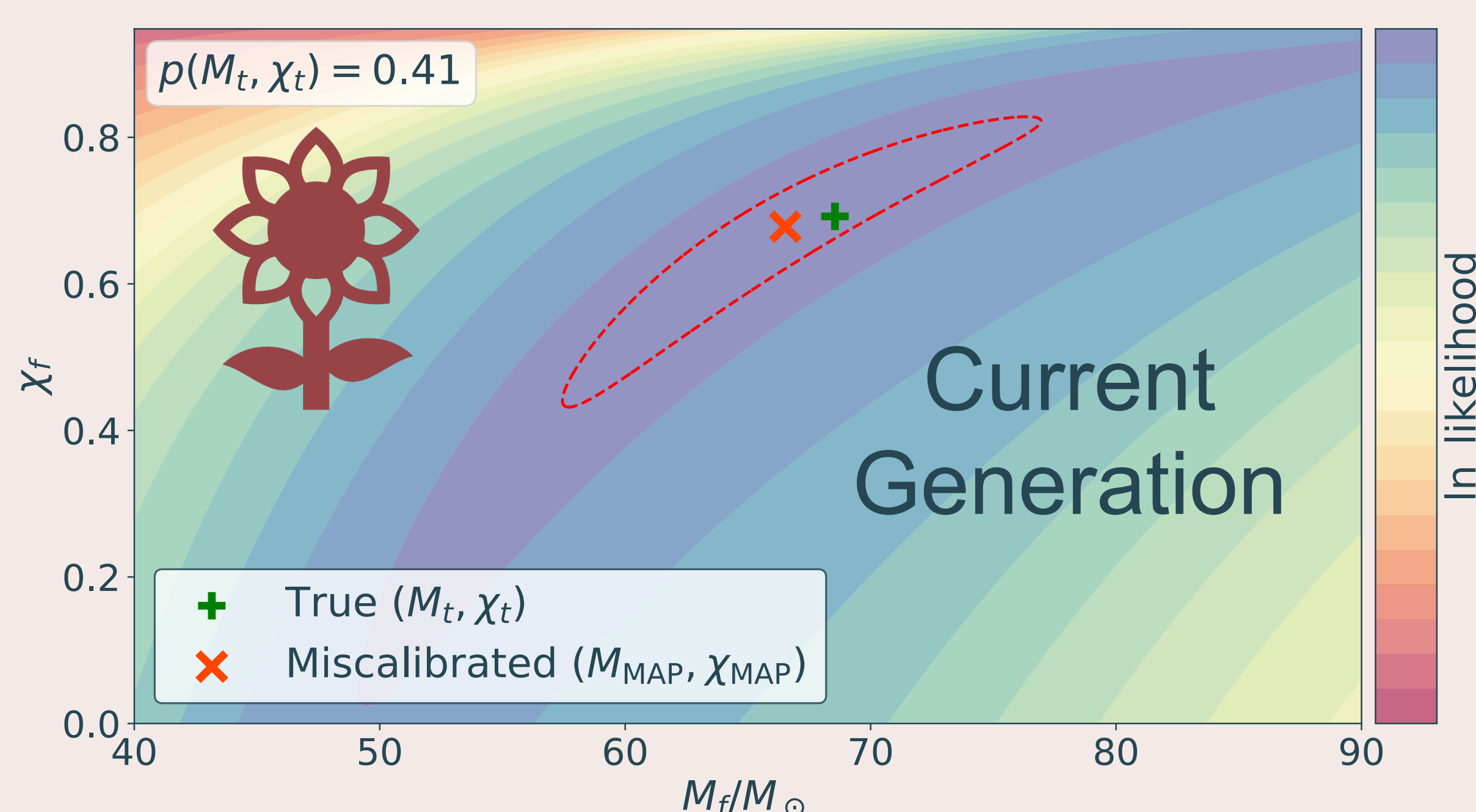
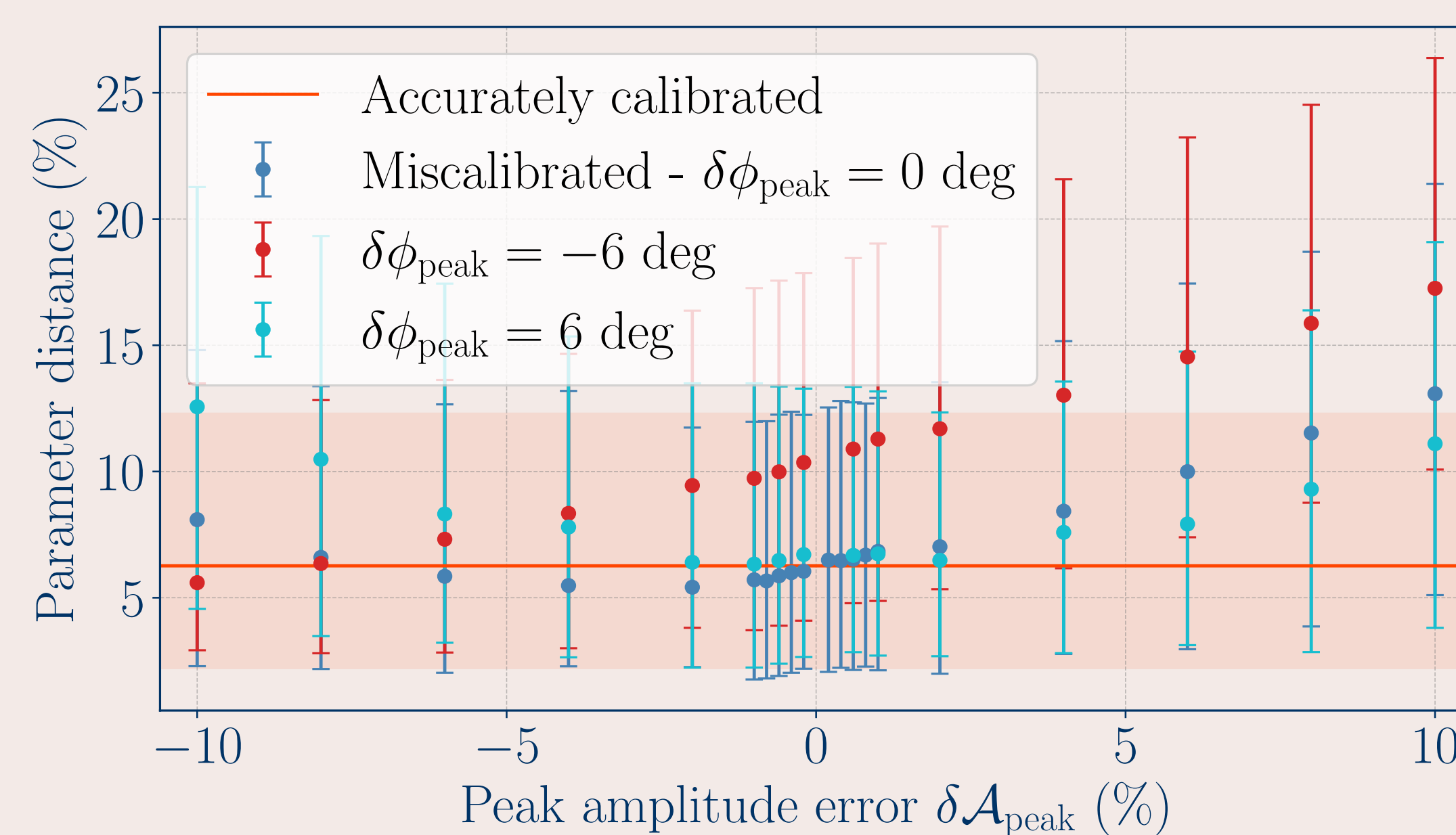
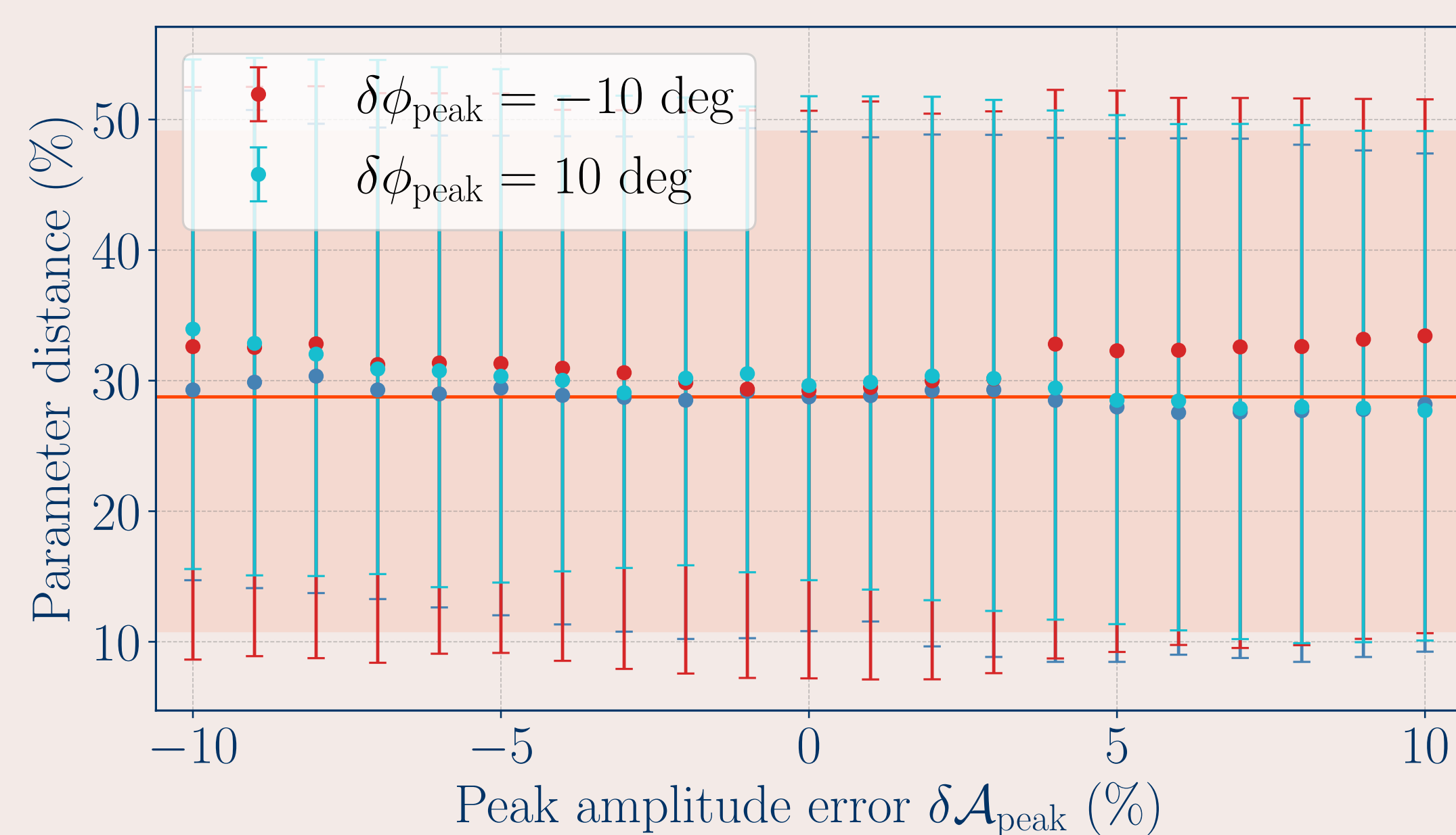


This time we
knew it was a
calibration
error...

BH spectroscopy aims to...

investigate the ringdown modes that present as a linear superposition in the signal post-merger (i.e., emitted by the remnant BH from a BBH merger).

They're generally unbiased for now, but with the next generation...



The current level of
calibration errors
($|\delta A| \leq 10\%$ & $|\delta \phi| \leq 10$ deg)
do not bias
analyses in the
current generation of
observatories, but we
will need improved
calibration accuracy
with the next
generation of
observatories.

